

Dataset-Grounded AI Copilot for Operational Decision Support

Final research-ready project proposal, architecture and implementation plan - v15

Purpose

This document is the final aligned project report for supervisor review and dissertation planning. It explains the research contribution, what changed after supervisor feedback, how the customer churn proof-of-concept works, how readiness and memory are handled, and how the system will be evaluated.

Item	Final position
Main case study	Customer churn decision support remains the primary proof-of-concept.
Research contribution	Readiness framework + evidence grounding + explainability-to-action + memory + evaluation.
Data integration	Local upload, URL/API, Google Drive and PostgreSQL structured data.
Memory architecture	Short-term session memory and ChromaDB long-term semantic memory. MongoDB is removed from the core design.
Proof evidence	Model metrics, readiness gates, Copilot evidence packages, audit logs, screenshots, exports and user evaluation.

1. What was completed before and what changed now

The previous code already contained a strong proof-of-concept base. The final version keeps those features and upgrades the research contribution, readiness criteria, memory architecture and evaluation framework.

Area	Previous work retained	Final v15 change
Data loading	CSV, Excel, JSON, ZIP, URL/API, Google Drive and PostgreSQL-ready loading	Kept and organised as an integrated data-source layer.
Visual analytics	Auto visual dashboard and manual graph builder	Kept; used as evidence-generation rather than decoration.
Modelling	Auto/manual classification and optional regression	Kept; model metrics exported as technical evidence.
Explainability	SHAP/fallback feature importance	Kept and connected to business recommendations.
Copilot	Dataset-grounded answer format	Upgraded with answer readiness, evidence package and memory retrieval.
MongoDB	Previously considered for unstructured storage	Removed from core architecture.
ChromaDB	Not central before	Added as semantic long-term memory for evidence retrieval.
Evaluation	Prediction-only vs explanation-supported comparison	Expanded with model, explanation, grounding, trust, usability and performance metrics.

2. Research background and gap

Current business intelligence tools, AutoML tools and AI assistants can generate dashboards, predictions or natural-language summaries. However, a dissertation-level gap remains around how a proof-of-concept Copilot can demonstrate that its answers are grounded in validated data, model evidence and explanation artefacts before supporting operational decisions.

- Many AI systems can produce confident answers, but the answer source and evidence path may not be visible to non-technical users.
- Explainable AI methods can identify model drivers, but they do not automatically translate explanations into safe business actions.
- Data quality problems such as missing values, schema mismatches, duplicates and weak targets can make Copilot answers unreliable if not checked first.
- A Copilot architecture should explain both short-term conversational memory and long-term semantic memory.
- The project therefore focuses on a customer churn proof-of-concept that validates data readiness, explains model outputs, generates evidence-backed recommendations and evaluates user trust and understanding.

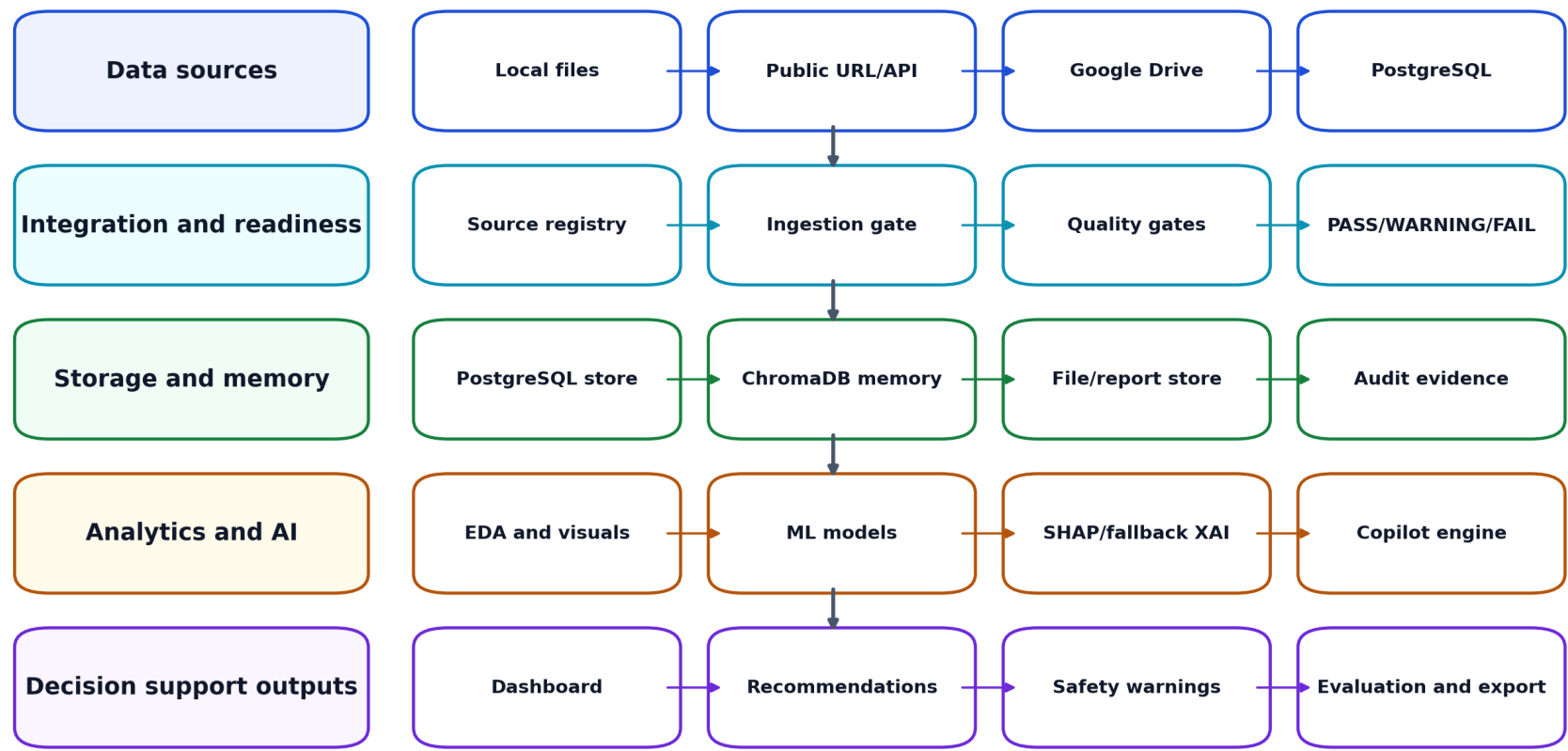
3. Research contribution framework

ID	Contribution	How it is implemented	Evidence in dissertation
C1	PASS/WARNING/FAIL readiness framework	Measurable gates for source, schema, missingness, duplicates, rows, target, class balance, privacy and question evidence.	Readiness table, screenshots, exported readiness report.
C2	Evidence-grounding mechanism	Evidence package combines source, columns, EDA statistics, model metrics, explanations, memory records, limitations and warnings.	Copilot response screenshots, audit logs, chat export.
C3	Explainability-to-action pipeline	SHAP/fallback drivers are translated into plain-English explanation, business recommendation and safety warning.	Model explanation output and recommendation table.
C4	Memory-supported Copilot architecture	Short-term session memory plus ChromaDB semantic memory for previous evidence and generated insights.	Memory page, retrieval examples and semantic memory export.
C5	Evaluation framework	Model, explanation, grounding, trust, usability, decision-support and performance metrics.	Questionnaire, charts, metric tables, user comments.

4. Integrated final architecture

Figure 1. Final Integrated System Architecture

Clean view of how data, readiness, storage, memory, modelling, explanation, Copilot answers and evaluation connect.



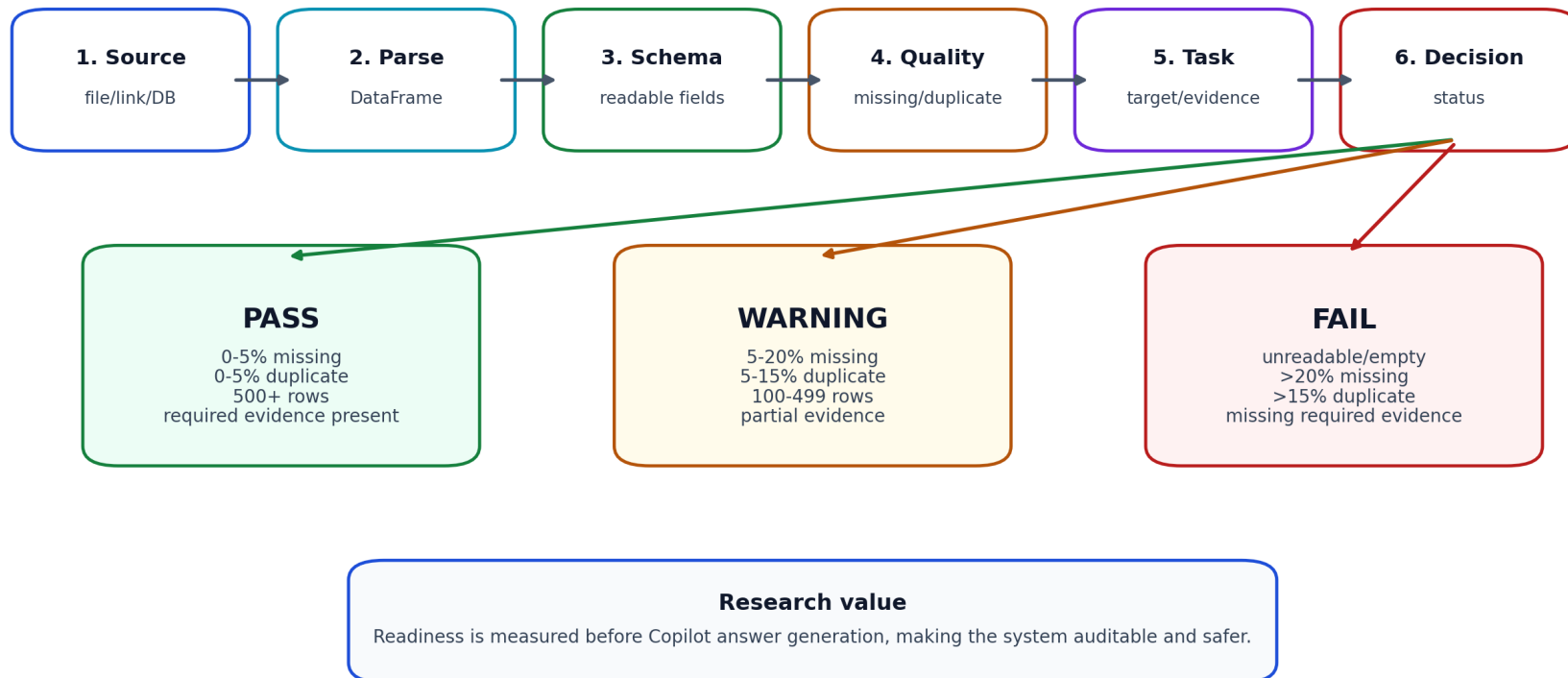
Main proof-of-concept: customer churn. Additional datasets: bank marketing, online retail/sales and customer feedback for robustness.

This figure shows how data moves from source connectors through readiness gates, storage, memory, analytics, explainability, Copilot response generation and the decision-support UI.

5. Readiness gates and data integration pipeline

Figure 2. Data Integration and Readiness Pipeline

Measurable gates decide whether the system answers, answers with caution, or refuses safely.

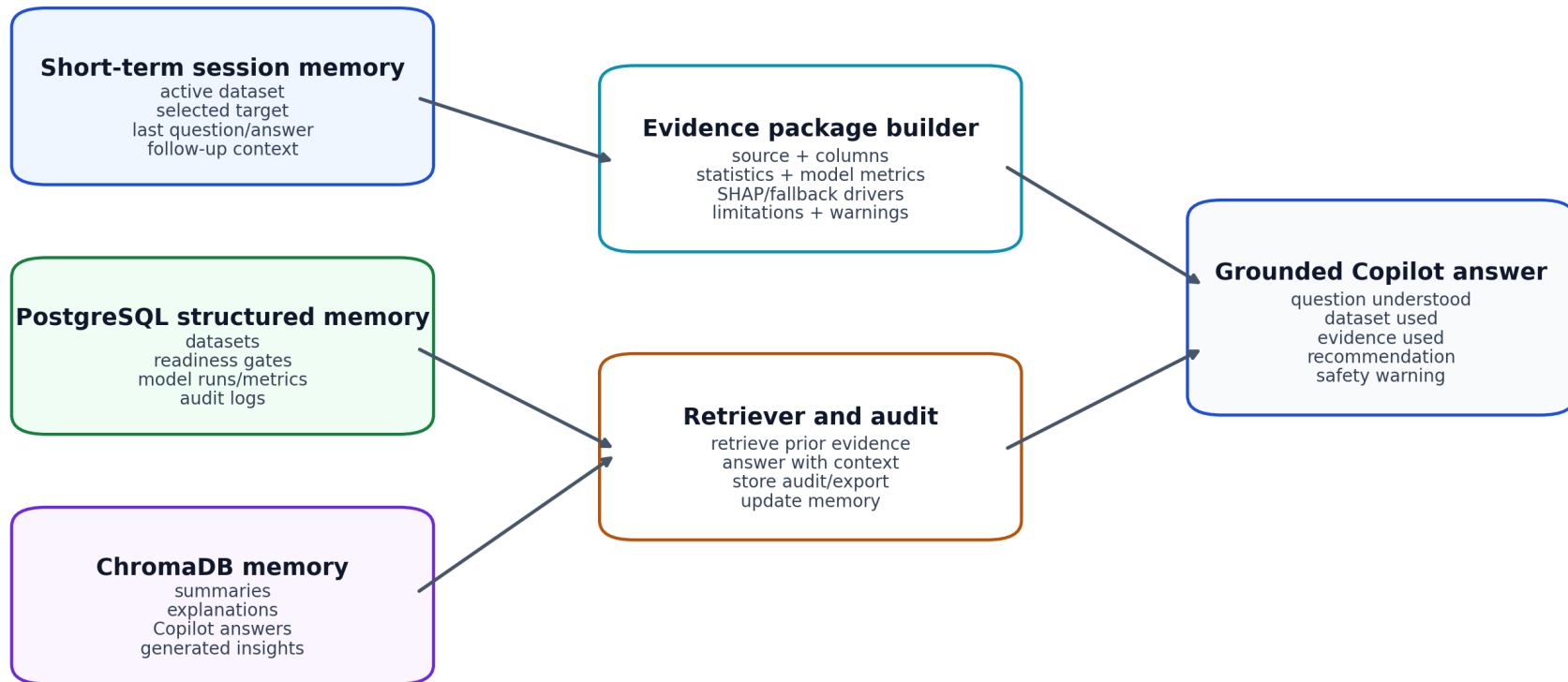


This figure defines how PASS, WARNING and FAIL are determined before analysis, modelling or answer generation.

6. Memory architecture

Figure 3. Memory Architecture: PostgreSQL + ChromaDB

The final design removes MongoDB from the core architecture and uses ChromaDB for semantic memory.



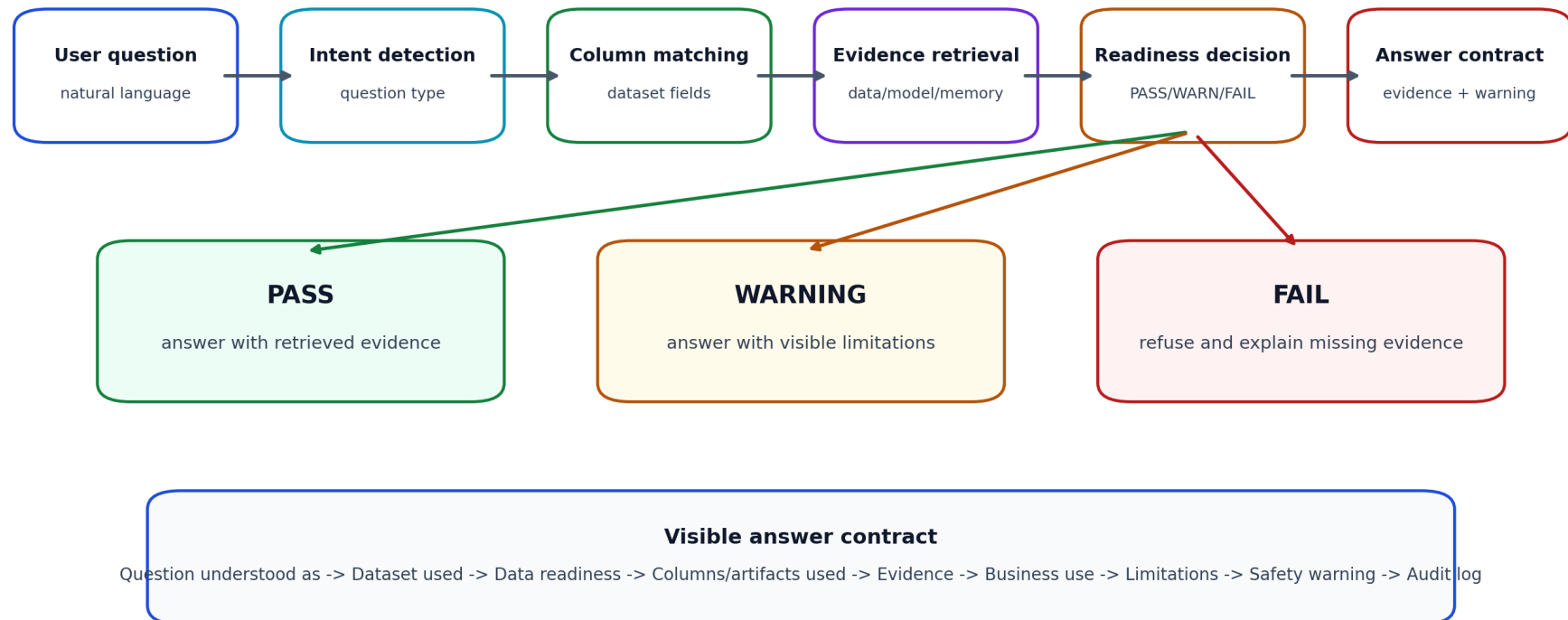
Why this matters: memory is explicit. Session memory supports follow-up questions; ChromaDB retrieves long-term evidence across sessions.

This figure clarifies short-term session memory, PostgreSQL structured storage and ChromaDB semantic memory. MongoDB is removed from the core design.

7. Copilot question processing and answer contract

Figure 4. Copilot Question Processing and Evidence Flow

The Copilot uses conditional logic for safe routing, but the answer is generated from active data and evidence.

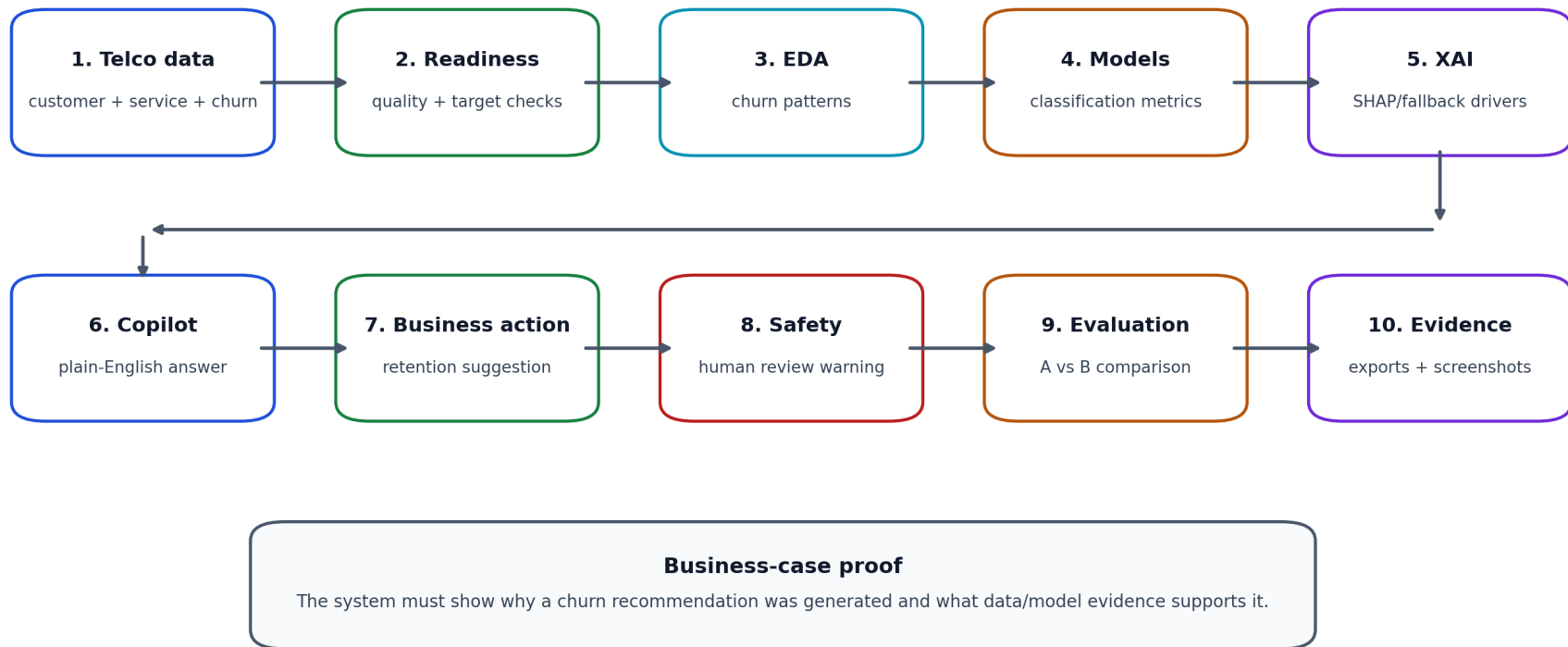


This figure explains where a Copilot answer comes from and why the system refuses or warns when evidence is missing.

8. Customer churn proof-of-concept workflow

Figure 5. Customer Churn Proof-of-Concept Workflow

Customer churn is the main implementation and evaluation case study.

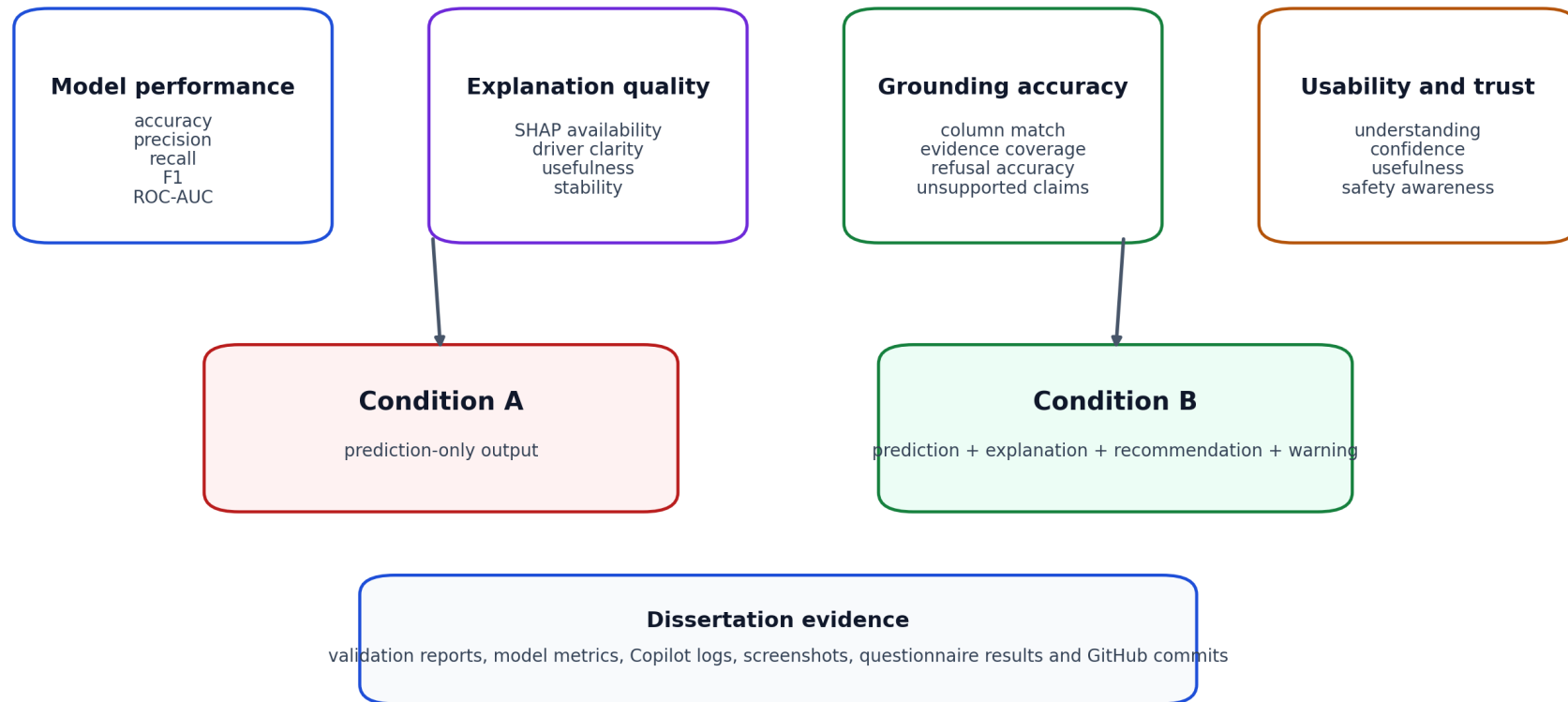


This figure connects the system architecture to the primary customer churn proof-of-concept case study.

9. Evaluation framework

Figure 6. Evaluation Framework

How the dissertation proves the prototype works.



This figure shows how the system will be proven using technical, explanation, grounding, trust and usability evidence.

10. Dataset plan and how each dataset is used

Dataset	Purpose	Expected target / key fields	How the Copilot should behave
Telco/customer churn	Main proof-of-concept and evaluation case study.	churn, tenure, contract, monthly charges, services, payment, optional feedback.	Predict churn, explain risk factors, recommend retention actions and warn that human review is required.
Bank marketing	Robustness test for campaign/response analysis.	y or response target, balance, age, job, campaign fields.	Answer campaign-response questions only when bank fields are active; refuse if using Telco data.
Online retail/sales	Large/medium data integration and visual analytics test.	Invoice date, product, quantity, unit price, customer/country.	Generate sales trends, top products, revenue evidence and stock/promotion suggestions.
Customer feedback	Text evidence and business improvement test.	feedback/comment/review text and optional churn/response link.	Find negative themes, evidence terms and improvement priorities; avoid unsupported sentiment claims.

11. PASS/WARNING/FAIL thresholds

Criterion	PASS	WARNING	FAIL
Missing cells	0-5% missing	>5-20% missing	>20% missing
Duplicate rows	0-5% duplicate	>5-15% duplicate	>15% duplicate
Rows for modelling	500+ rows	100-499 rows	<100 rows
Class balance	Minority class $\geq 10\%$	Minority class 5-10%	Minority class $< 5\%$
Target readiness	Target present with 2+ values/classes	Target needs confirmation or weak balance	Target missing, constant or unusable
Question evidence	Required columns and evidence available	Partial evidence or approximate column match	Required column/evidence missing
Security/privacy	No obvious sensitive fields	Possible PII detected	High-risk sensitive data or unauthorised access

The thresholds are intentionally transparent so the readiness framework can be described and evaluated in the dissertation. They are not a hidden black-box score.

12. Modelling, accuracy and explainability

The prototype is classification-first because customer churn is the primary case study. Regression remains optional for numeric business outcomes. Automatic modelling runs once after upload when a suitable target is detected, and manual retraining remains available for controlled testing.

Component	Implementation
Classification models	Logistic Regression, Random Forest, Extra Trees, Gradient Boosting and optional XGBoost.
Regression models	Linear Regression, Random Forest Regressor, Extra Trees Regressor, Gradient Boosting Regressor and optional XGBoost Regressor.
Model selection evidence	Accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix and training rows.
Why F1/recall matter	Churn datasets are often imbalanced; accuracy alone can hide poor detection of churners.
Explainability	SHAP where supported; fallback feature importance or model coefficients if SHAP is unavailable.
Limitations	Metrics are prototype evidence, not automatic business approval. Human review is required before action.

13. Copilot answer format

Every important Copilot answer should use a visible answer contract. This makes the answer auditable and helps the user understand whether the output is supported by data.

Answer part	Purpose
Question understood as	Shows how the system interpreted the user request.
Dataset used	Shows active file/source, rows and columns.
Data readiness	Shows PASS/WARNING/FAIL and quality score.
Columns/artefacts used	Shows the data fields, model results, explanations or memory records used.
Evidence from integrated data	Gives the actual statistical/model/text evidence.
Business/decision-support use	Explains how the evidence can support action.
Limitations	States missing context and scope boundaries.
Safety warning	Reminds that output is decision support and requires human review.

14. Implementation file and script structure

File/folder	Purpose
app.py	Main application, navigation, dashboard, model page, chat, evaluation, export and research/memory page.
src/data_loader.py	Local upload, URL/API, Google Drive and PostgreSQL loading.
src/cleaning.py	Automatic column cleaning, type handling and cleaning report.
src/validation.py + src/readiness_gates.py	Measurable readiness report and PASS/WARNING/FAIL rules.
src/modeling.py	Classification and regression model training and metrics.
src/explainability.py	SHAP/fallback explanations and top-driver evidence.
src/copilot.py	Controlled dataset-grounded intent routing and answer generation.
src/evidence_package.py	Creates traceable evidence package for answers and memory.
src/memory_manager.py	Short-term memory and memory architecture table.
src/vector_memory.py	ChromaDB semantic memory with JSONL fallback.
src/business_insights.py	Dataset-grounded business improvement suggestions.
src/visual_analytics.py	Automatic visual dashboard and manual chart builder.
tests/	Regression tests for cleaning, modelling, Copilot, visual analytics and v15 research/memory behaviour.

15. Evaluation plan and proof that it works

Evaluation dimension	Metrics/evidence
Technical/model performance	Accuracy, precision, recall, F1, ROC-AUC, confusion matrix, runtime and rows processed.
Explanation quality	SHAP/fallback availability, top-driver coverage, clarity and usefulness ratings.
Grounding accuracy	Required-column match, evidence-source availability, unsupported-claim rate and refusal accuracy.
Trustworthiness	Evidence coverage, limitation coverage, safety-warning coverage and audit/export consistency.
Usability	Likert scores, task completion, perceived usefulness and comments.
Decision support	Condition A: prediction-only vs Condition B: prediction + explanation + recommendation + warning.

The dissertation should include screenshots of data upload, readiness PASS/WARNING/FAIL, model metrics, explanations, Copilot evidence answers, Copilot refusal answers, business recommendations, memory records, exports and evaluation responses.

16. Security, cloud readiness and TRL

Area	Final position
Security	No hard-coded secrets, environment-variable credentials, file validation, PII warning, human-review warnings and no automatic high-risk decisions.
LLM boundary	The implemented prototype is controlled and dataset-grounded. Any future LLM layer must receive a restricted evidence package and must not directly query raw databases.
Cloud implementation	Prototype is local. Scalable design separates UI, API/backend, PostgreSQL, ChromaDB, model service, explanation service, audit logs and monitoring. Google Cloud can be discussed as deployment option.
TRL now	TRL 3-4: proof-of-concept demonstrated in a local/lab environment.
TRL target	TRL 4-5 if validated with realistic public datasets, user evaluation and documented evidence.

17. Final demonstration script

- Upload the Telco churn dataset and show the active source and readiness status.
- Open Cleaning & EDA and show missing values, duplicates and target distribution.
- Open Modelling & Prediction Explanation and show automatic model status, model metrics and SHAP/fallback drivers.
- Ask the Copilot: "Why are customers at high churn risk and what should we improve?"
- Ask a fail-case question such as "Does balance affect campaign response?" while Telco data is active and show that the system refuses because balance is not present.
- Open Research & Memory and show contribution table, readiness criteria, short-term memory and ChromaDB/JSONL memory status.
- Export the report, chat log, readiness report and audit evidence.

Final message for supervisor

The final prototype is a customer churn proof-of-concept that demonstrates integrated data ingestion, measurable readiness gates, explainable modelling, evidence-grounded Copilot answers, business recommendations, memory support and evaluation evidence. The contribution is the integrated research framework and proof-of-concept, not a new ML algorithm or a commercial BI replacement.